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```
% Justin Pulley
% EE 362
% Project 1
```

```
clear all
close all
clc
```

```
% Units
k = 1e3; % kilo
m = 1e-3; % milli
u = 1e-6; % mu
```

```
% Global Variables
f = logspace(1,5,1000); % frequency
w = 2*pi*f; % radian frequency
```

## Example a) High Pass

```
R = 100; % Resistor Value
L = 10*m; % Inductor Value
imp_L = j*w*L; % Impedance of Inductor

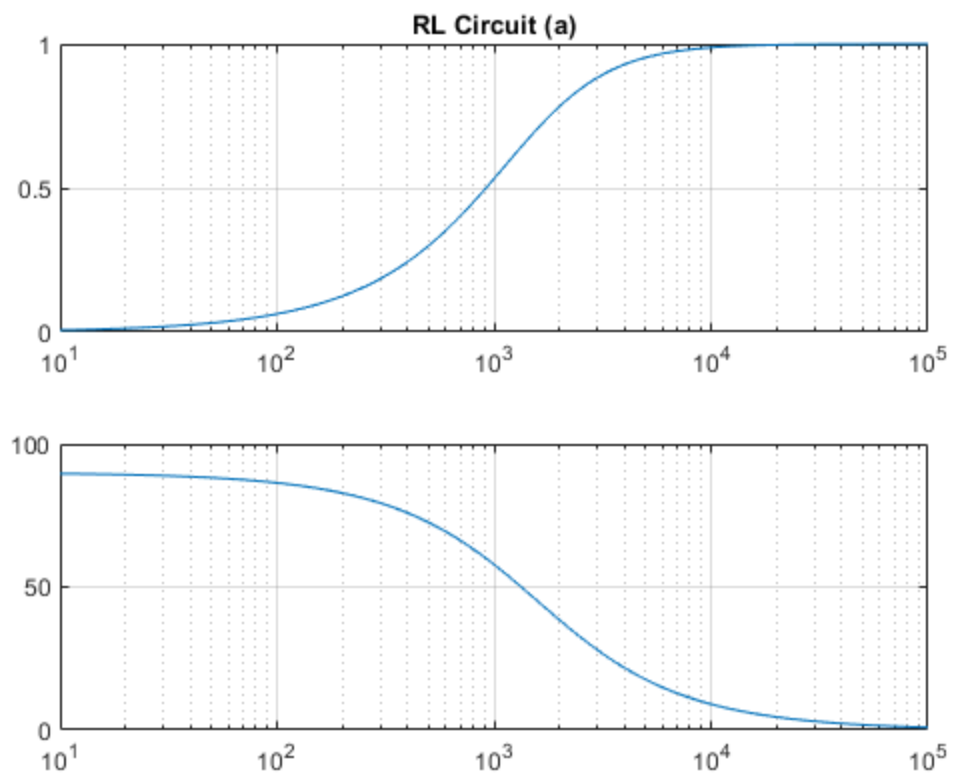
H_jw = imp_L./(R + imp_L); % Transfer Function

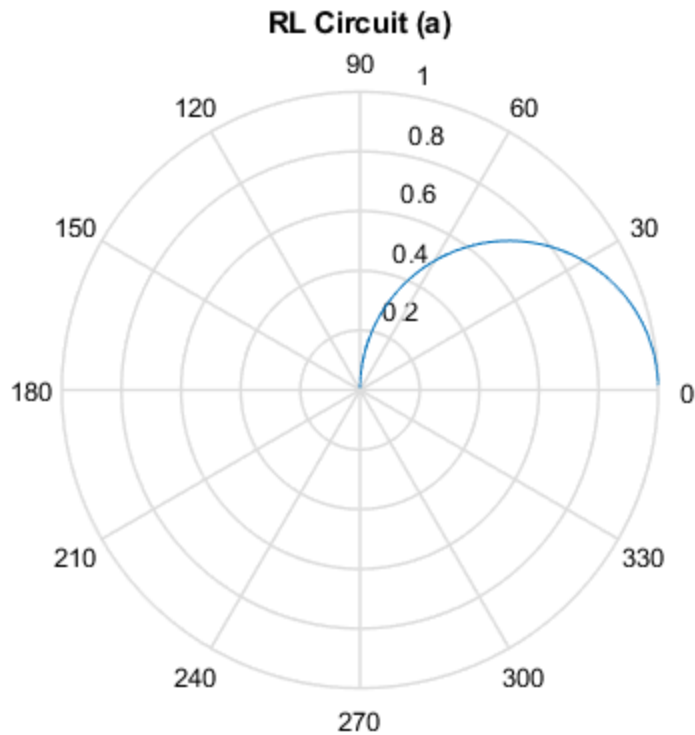
figure;
subplot(211);
semilogx(f,abs(H_jw)); % gain vs frequency H_jw
grid
title('RL Circuit (a)');
subplot(212);
semilogx(f,angle(H_jw)*180/pi); % phase-shift vs frequency H_jw
grid

figure;
polar(angle(H_jw),abs(H_jw)); % polar plot of H_jw
grid
```

---

```
title('RL Circuit (a)');
```





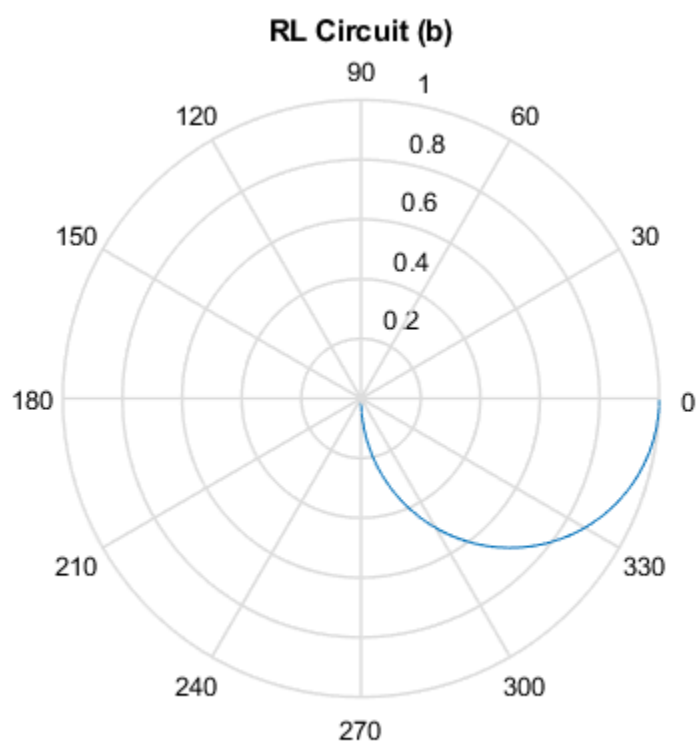
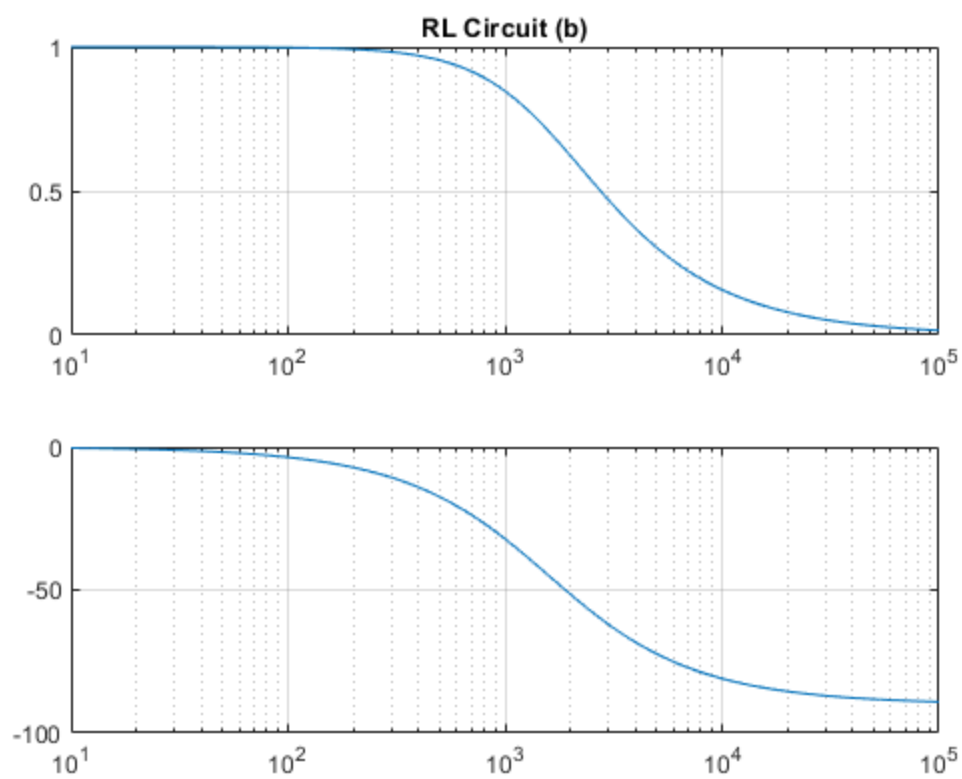
## b) Low Pass

```
R = 100; % Resistor Value
L = 10*m; % Inductor Value
imp_L = j*w*L; % Impedance of Inductor

H_jw = R./(R + imp_L); % Transfer Function

figure;
subplot(211);
semilogx(f,abs(H_jw)); % gain vs frequency H_jw
grid
title('RL Circuit (b)');
subplot(212);
semilogx(f,angle(H_jw)*180/pi); % phase-shift vs frequency H_jw
grid

figure;
polar(angle(H_jw),abs(H_jw)); % polar plot of H_jw
grid
title('RL Circuit (b)');
```



---

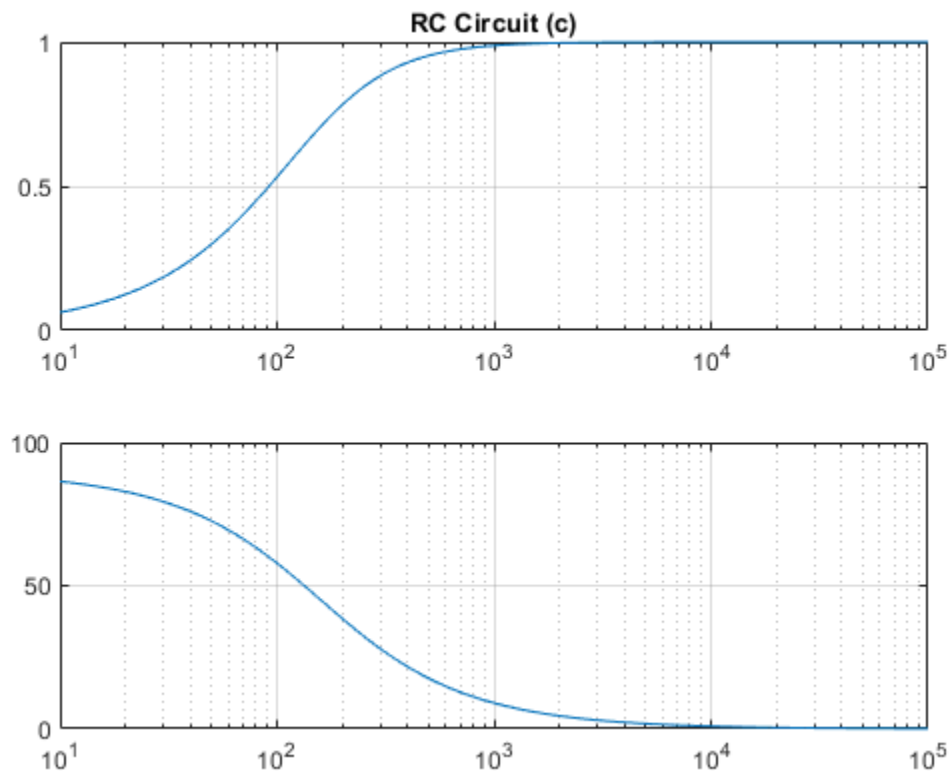
## c) High Pass

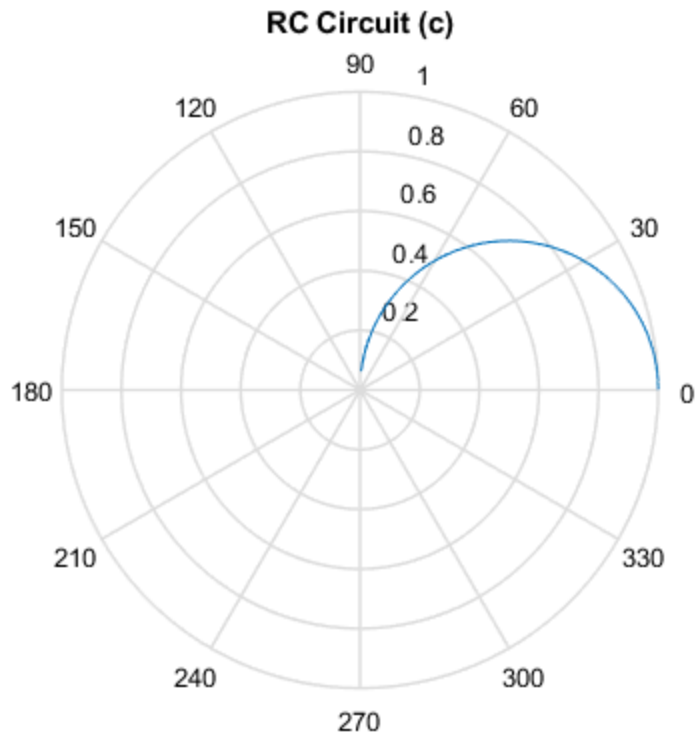
```
R = 1*k; % Resistor Value
C = 1*u; % Capacitor Value
imp_C = 1./(j*w*C); % Impedance of Capacitor

H_jw = R./(R + imp_C); % Transfer Function

figure;
subplot(211);
semilogx(f,abs(H_jw)); % gain vs frequency H_jw
grid
title('RC Circuit (c)');
subplot(212);
semilogx(f,angle(H_jw)*180/pi); % phase-shift vs frequency H_jw
grid

figure;
polar(angle(H_jw),abs(H_jw)); % polar plot of H_jw
grid
title('RC Circuit (c)');
```





## d) Low Pass

```

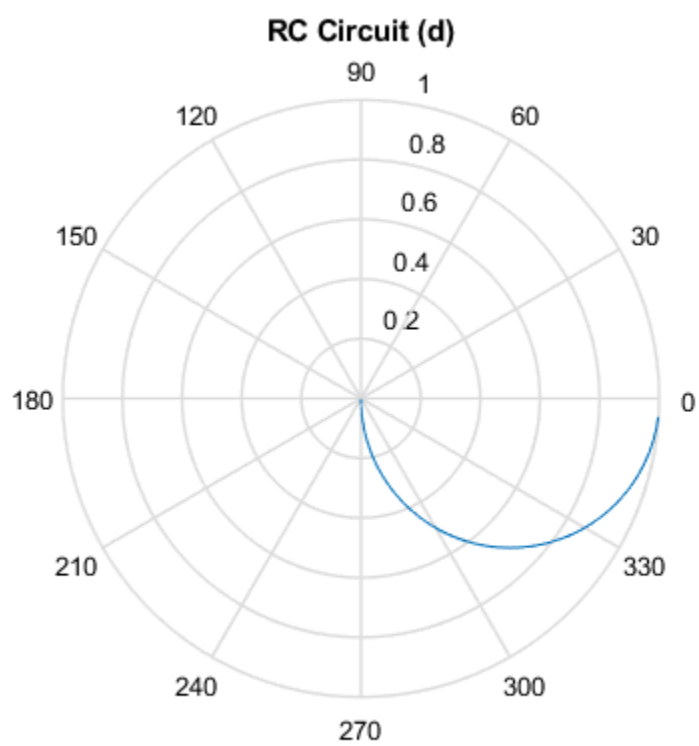
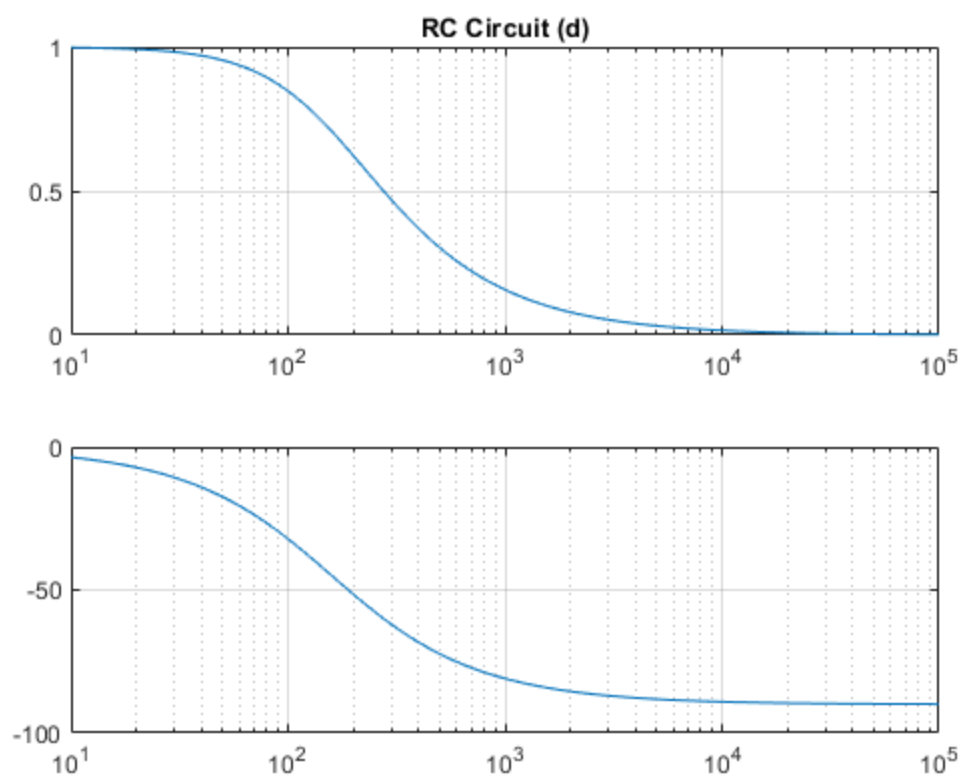
R = 1*k; % Resistor Value
C = 1*u; % Capacitor Value
imp_C = 1./(j*w*C); % Impedance of Capacitor

H_jw = imp_C./(R + imp_C); % Transfer Function

figure;
subplot(211);
semilogx(f,abs(H_jw)); % gain vs frequency H_jw
grid
title('RC Circuit (d)');
subplot(212);
semilogx(f,angle(H_jw)*180/pi); % phase-shift vs frequency H_jw
grid

figure;
polar(angle(H_jw),abs(H_jw)); % polar plot of H_jw
grid
title('RC Circuit (d)');

```



---

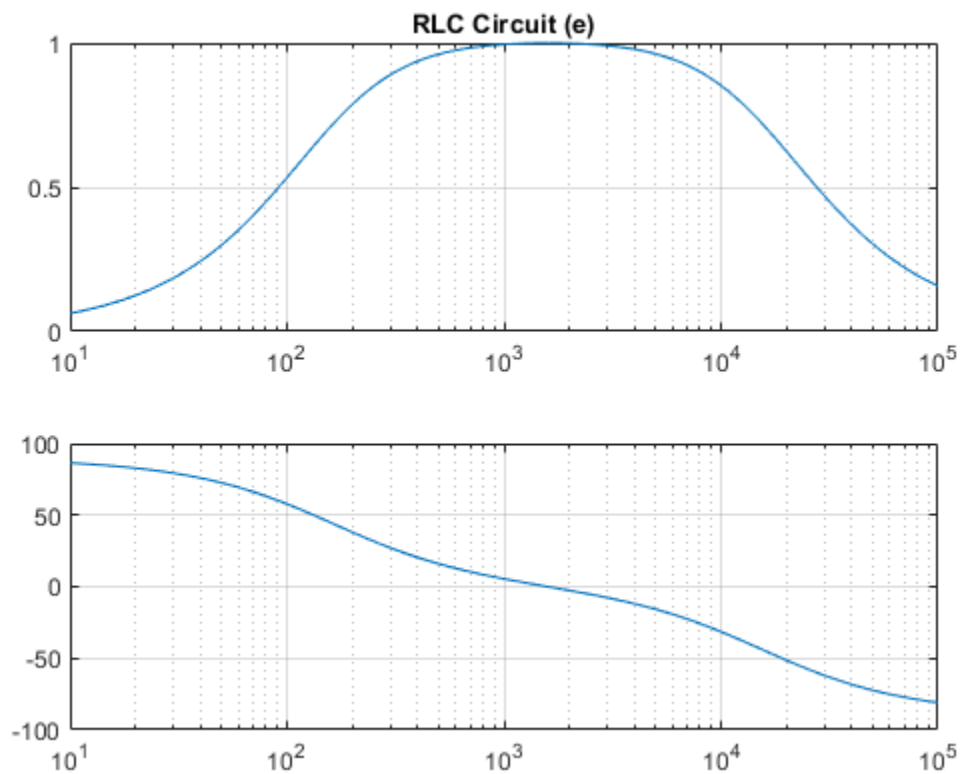
## e) Band Pass Filter

```
R = 1*k; % Resistor Value
C = 1*u; % Capacitor Value
L = 10*m; % Inductor Value
imp_L = j*w*L; % Impedance of Inductor
imp_C = 1./(j*w*C); % Impedance of Capacitor

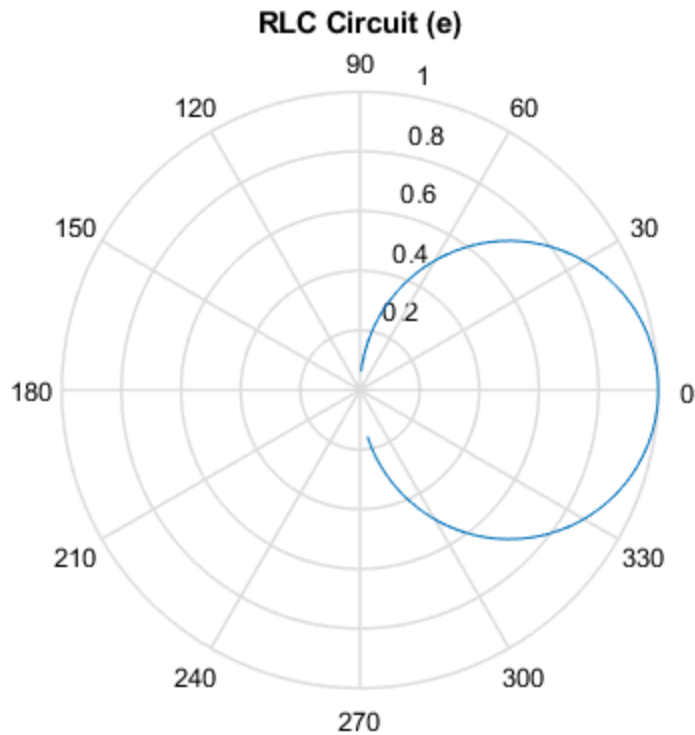
H_jw = R./(R+imp_L+imp_C); % Transfer Function

figure;
subplot(211);
semilogx(f,abs(H_jw)); % gain vs frequency H_jw
grid
title('RLC Circuit (e)');
subplot(212);
semilogx(f,angle(H_jw)*180/pi); % phase-shift vs frequency H_jw
grid

figure;
polar(angle(H_jw),abs(H_jw)); % polar plot of H_jw
grid
title('RLC Circuit (e)');
```







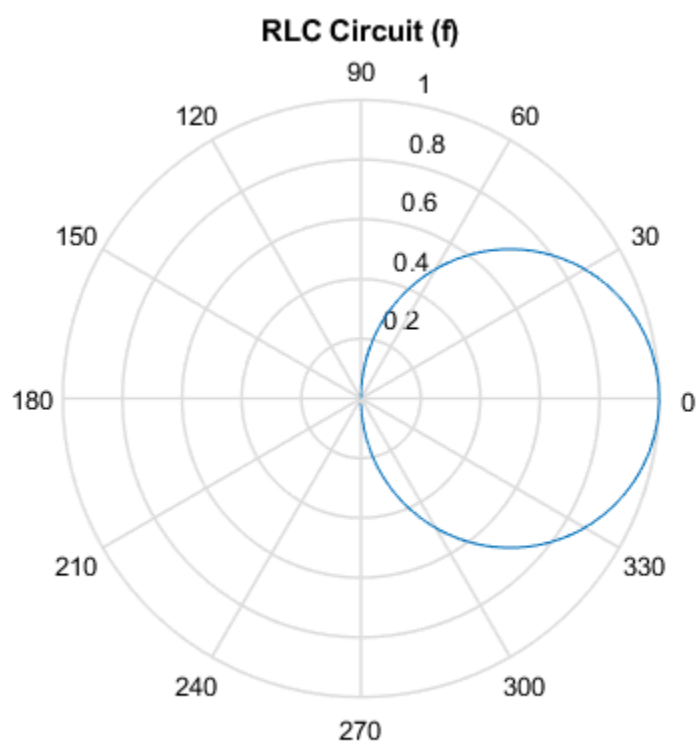
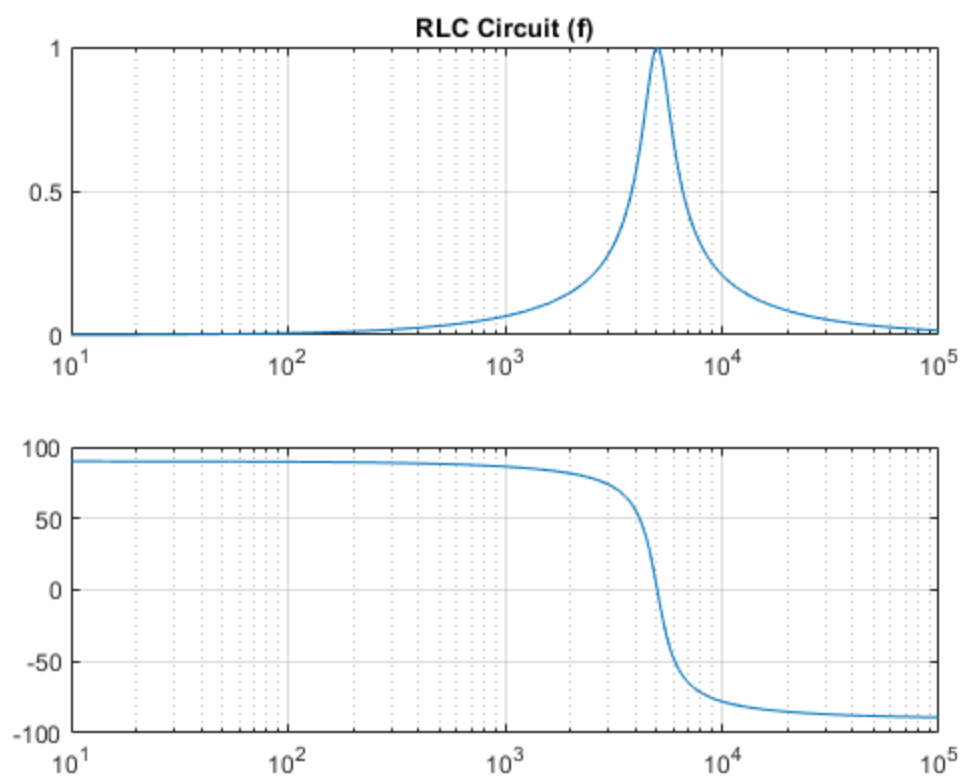
## f) Band Pass Filter

```
R = 100; % Resistor Value
C = 1*u; % Capacitor Value
L = 1*m; % Inductor Value
imp_L = j*w*L; % Impedance of Inductor
imp_C = 1./(j*w*C); % Impedance of Capacitor
LPC = (imp_L.*imp_C)./(imp_L+imp_C); % Impedance of Inductor in Parallel w/
Capacitor
```

```
H_jw = LPC./(R+LPC); % Transfer Function
```

```
figure;
subplot(211);
semilogx(f,abs(H_jw)); % gain vs frequency H_jw
grid
title('RLC Circuit (f)');
subplot(212);
semilogx(f,angle(H_jw)*180/pi); % phase-shift vs frequency H_jw
grid
```

```
figure;
polar(angle(H_jw),abs(H_jw)); % polar plot of H_jw
grid
title('RLC Circuit (f)');
```



---

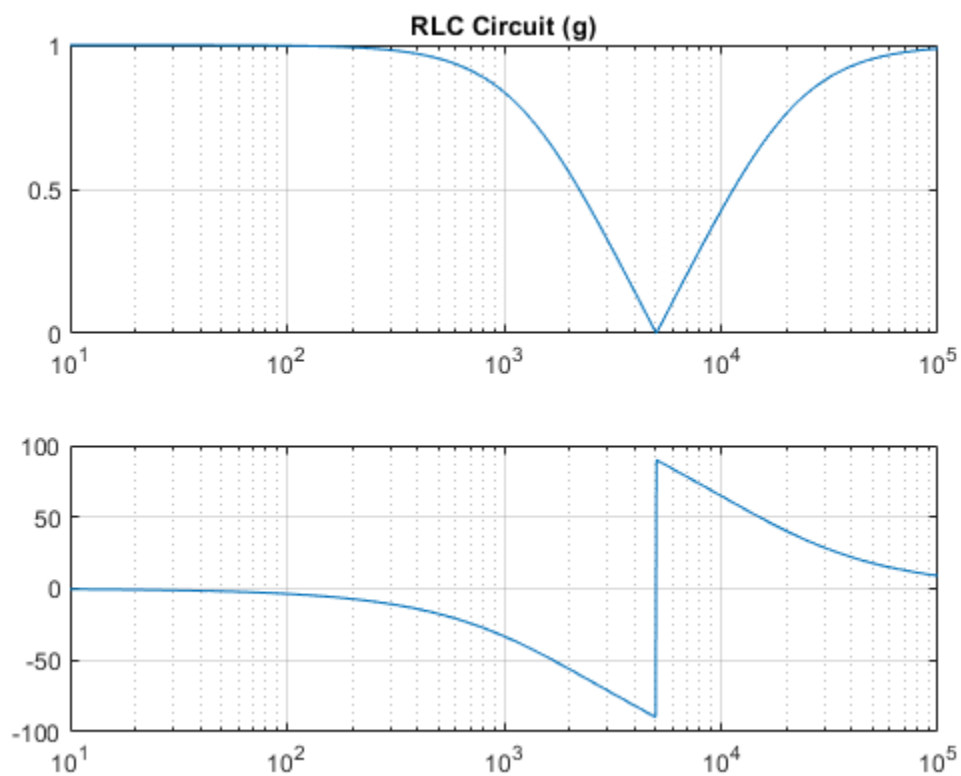
## g) Notch Filter

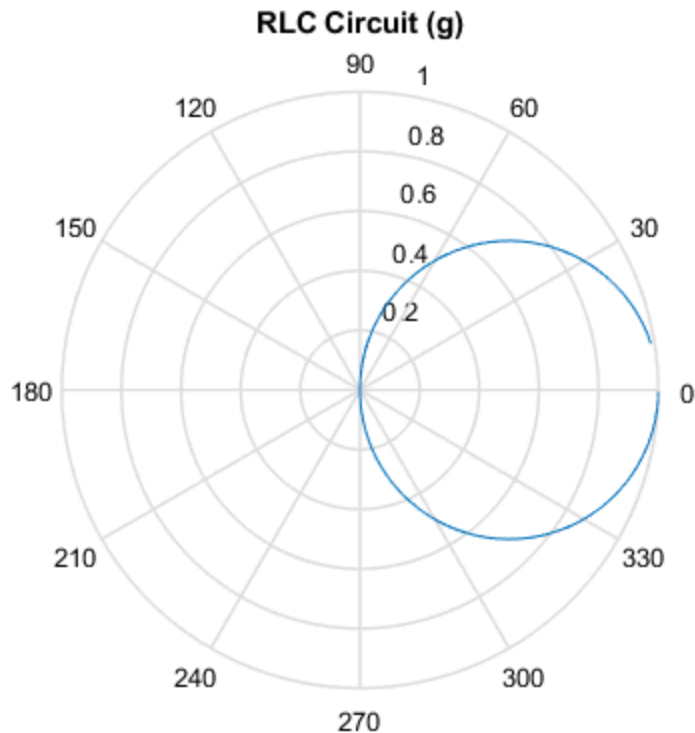
```
R = 100; % Resistor Value
C = 1*u; % Capacitor Value
L = 1*m; % Inductor Value
imp_L = j*w*L; % Impedance of Inductor
imp_C = 1./(j*w*C); % Impedance of Capacitor

H_jw = (imp_C+imp_L)./(R+imp_C+imp_L); % Transfer Function

figure;
subplot(211);
semilogx(f,abs(H_jw)); % gain vs frequency H_jw
grid
title('RLC Circuit (g)');
subplot(212);
semilogx(f,angle(H_jw)*180/pi); % phase-shift vs frequency H_jw
grid

figure;
polar(angle(H_jw),abs(H_jw)); % polar plot of H_jw
grid
title('RLC Circuit (g)');
```





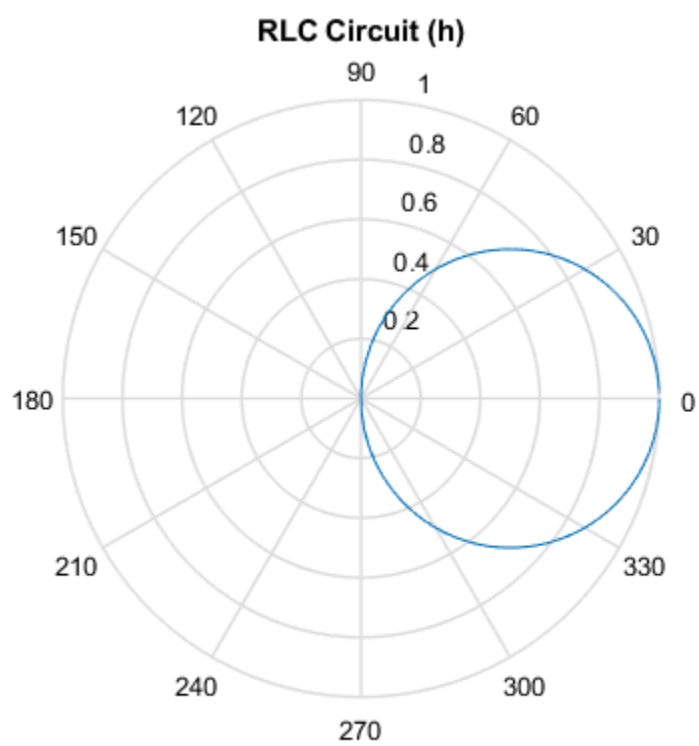
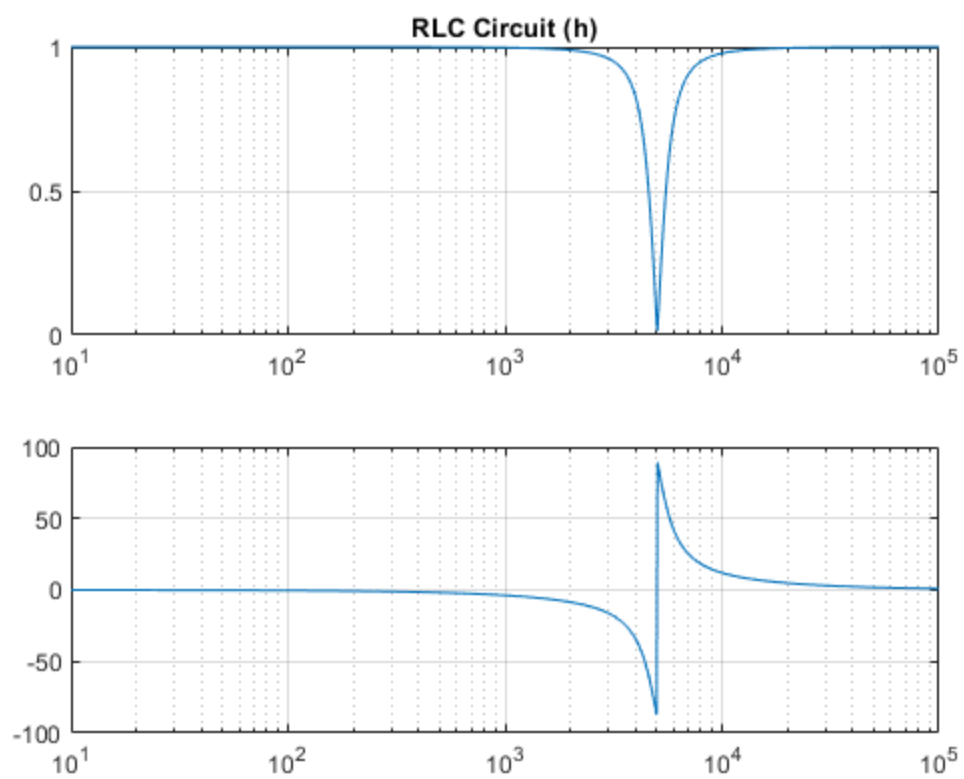
## h) Notch Filter

```
R = 100; % Resistor Value
C = 1*u; % Capacitor Value
L = 1*m; % Inductor Value
imp_L = j*w*L; % Impedance of Inductor
imp_C = 1./(j*w*C); % Impedance of Capacitor
LPC = (imp_L.*imp_C)./(imp_L+imp_C); % Impedance of Inductor in Parallel w/
Capacitor
```

```
H_jw = R./(R+LPC); % Transfer Function
```

```
figure;
subplot(211);
semilogx(f,abs(H_jw)); % gain vs frequency H_jw
grid
title('RLC Circuit (h)');
subplot(212);
semilogx(f,angle(H_jw)*180/pi); % phase-shift vs frequency H_jw
grid
```

```
figure;
polar(angle(H_jw),abs(H_jw)); % polar plot of H_jw
grid
title('RLC Circuit (h)');
```



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